WONJOON HWANG

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SUMMARY

M.S. in Data Science candidate and Databricks Student Fellow with experience building ML and data workflows using Python, SQL, Spark, and Databricks. Research and project experience spans federated ML, predictive modeling, anomaly detection, model evaluation, and end-to-end data pipelines across environmental and geospatial datasets.

EDUCATION

UNIVERSITY OF WASHINGTON **Sep 2025 – Mar 2027**
M.S., Data Science (GPA: 3.86) *Seattle, WA*

UNIVERSITY OF GEORGIA **Aug 2022 – May 2025**
B.S., Data Science | Minors: Statistics & Computer Science (Major GPA: 3.68) *Athens, GA*
- **Achievements:** Dean's List: Spring 2023, Fall 2023, Spring 2024

EXPERIENCE

Databricks | *Databricks Student Fellow* **Jun 2026 – Present**
- Build end-to-end **Python, SQL, Spark, and Databricks** workflows for ingestion, transformation, validation, analytics, and ML.
- Develop reusable technical demos of lakehouse and data-engineering workflows for mixed technical audiences.
- Selected as a **Databricks Student Fellow** after a **1st-place UW hackathon** finish; presented TerraCast at the Data + AI Summit.

Translation Commons | *Volunteer Data Engineer* **Jun 2026 – Sep 2026**
- Shipped **21 production fixes** across **9 language groups**, correcting parent-child relationships and improving shared dataset consistency.
- Built a scope-validation tool that detected **95 hierarchy inconsistencies**, catching structural data issues before downstream use.
- Added **9 unit tests** for parent-override behavior and issue detection to validate future hierarchy updates.

University of Georgia | *Research Assistant - Federated & Distributed ML* **Sep 2023 – Jan 2024**
- Implemented and evaluated **FedSVD and FedPower** in **Python and TensorFlow**, simulating decentralized ML across multiple clients with **non-IID data** to study performance under heterogeneous distributions.
- Benchmarked federated methods against **centralized and baseline approaches**; analyzed **convergence, accuracy, and communication/computation trade-offs** to compare distributed training strategies.

PROJECTS

TerraCast **Mar 2026**
Databricks × UW Paul G. Allen School Hackathon - 1st Place
- Presented TerraCast's crop-yield and regional-risk analytics at the **Data + AI Summit 2026** after our team earned **1st place** at the Databricks × UW Hackathon.
- Built a **Databricks** pipeline integrating crop, weather, and climate data with **Python, SQL, and Spark**; a **LightGBM-Random Forest ensemble** achieved **0.88 mean R^2** under **historical county-held-out 5-fold CV**.
- Contributed to an audit tracing a **pandas merge/index-alignment bug** affecting stored predictions and anomaly counts, while confirming fold-level CV R^2 was unaffected.

Water Quality Prediction **Jan 2026 – Mar 2026**
EY Open Science AI & Data Challenge, Snowflake Collaboration
- Integrated and validated **9,319 water-quality observations** with **205 satellite, climate, hydrology, and geospatial features** from Landsat, TerraClimate, ERA5, and HydroRIVERS.
- Engineered environmental features and designed **spatial cross-validation** to reduce leakage and evaluate **generalization across unseen regions**.
- Ranked **66th of 2,500+ teams** on the final leaderboard in the 2026 EY Open Science AI & Data Challenge.

Darin **Jun 2026 – Present**
Founder / Solo Developer | iOS TestFlight Beta
- Founded and built Darin, an **iOS baby-care journal in TestFlight beta**, using **React Native, TypeScript, and Supabase**.
- Built voice-assisted care logs, photo diaries, growth books, family roles, private media access, and a **5-language UI**.

WatchDawg **Sep 2025 – Dec 2025**
University of Washington
- Developed **Python and SQL** workflows with **Dash and DuckDB** to transform, aggregate, and visualize public-safety data.
- Built interactive **geospatial and time-series views** to explore incident patterns by location, time, precinct, and incident type.
- Added **LightGBM Poisson forecasting** and drill-down analysis to surface changes in local incident patterns.

SKILLS

- **Programming:** Python, SQL, R, Java, TypeScript
- **ML/Statistics:** TensorFlow, PyTorch, scikit-learn, LightGBM, XGBoost, federated ML, model evaluation, statistical modeling
- **Data & Analytics:** Spark, PySpark, Databricks, Pandas, NumPy, ETL/data pipelines, data validation, geospatial analysis, anomaly detection, forecasting
- **Databases:** PostgreSQL, MySQL, DuckDB, Snowflake, Supabase
- **Cloud & Systems:** AWS, Google Cloud, Git, Docker, REST APIs, CI/CD, backend systems, automation workflows